# Project INSPIRE Design Document

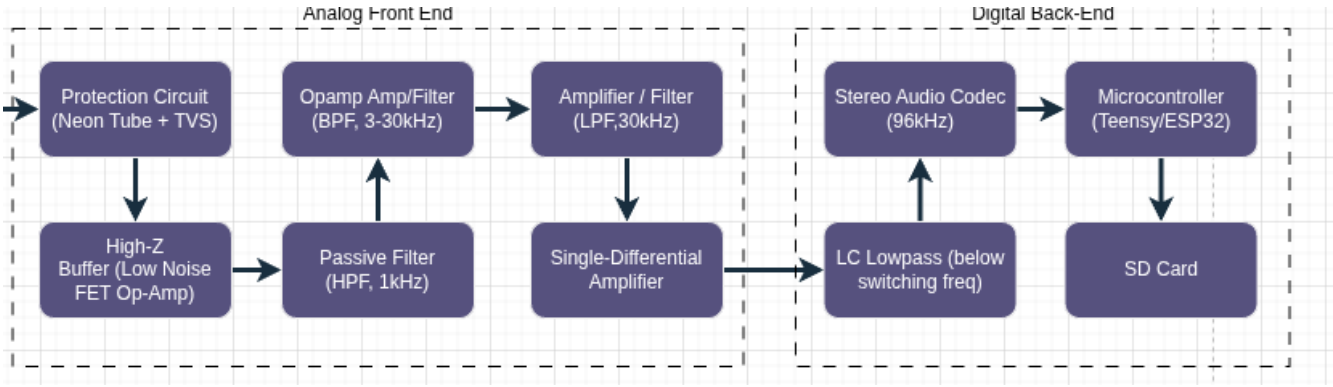
## Project Part II: High-Z Buffer

Revision 1.0: Presented 10/9/2026

## 1) General Project Specifications

The purpose of the project is to build a receiver to capture natural radio atmospheric phenomena in the Very Low Frequency (VLF) range of 3-30kHz.

A block diagram of the project is provided below:




The project will be built as two Printed Circuit Boards (PCBs) inside shielded metal enclosures, and powered off batteries, in order to establish isolation and distance from power lines, which are very large emitters of interference.

In order to pick up these signals, a high resolution 24-bit audio Analog-Digital Converter (ADC) is required. As such, we can establish a dynamic range of approximately 100dB. This means that the analog stage should be prioritized to ensure linearity and prevent clipping.

## 2) Sub-circuit Specifications

The High-Z Buffer is critical because the E-field whip antenna acts as an extremely small value capacitor (15-20pF). To prevent signal attenuation and loading effects, the input stage must present an extremely high input impedance (High-Z).

This sub-circuit will utilize a Low-Noise Junction FET (JFET) or CMOS Operational Amplifier configured as a voltage follower, or unity-gain noninverting amplifier. This design must minimize noise and input capacitance, while operating cleanly within the system's power limits.

- Active Element Selection: A low-noise FET-input op-amp must be selected. Because FET inputs possess inherently lower input bias currents, they minimize current noise and prevent loading down the antenna signal.
- The selected op-amp must operate reliably on the existing dual-3.3V analog power rails.
- Key datasheet parameters to prioritize include Input Bias Current, Input Capacitance, and Voltage Noise Density.
- Biasing and Stability: A high-value resistor, typically 10M or higher, must be placed from the non-inverting input to ground to provide a DC path for the input bias current. This prevents the op-amp output from drifting to the supply rails.
- Bandwidth & Linearity: The buffer must remain perfectly linear across the entire 3–30kHz VLF range without introducing distortion or clipping. This will ensure a clearer signal at the Analog-Digital Converter.

## 3) Expected Outputs

This circuit design requires a writeup and simulation with three major parts:

- **Op-Amp Selection & Justification**: A selected model of a low-noise, FET-input operational amplifier complete with a parameter

table (Gain Bandwidth Product, Slew Rate, Input Bias Current, Noise Density) and a short paragraph with justification.

- **DC Biasing Network Design:** A small set of calculations for the and justification of the input biasing resistors, which helps ensure proper input impedance requirements are met without injecting excessive thermal noise.
- **SPICE Simulation:** A spice simulation should be undertaken, with component model imported, and two major simulation runs should be run:
  - An AC sweep analysis from below our band to above it must be done, which shows that the circuit is not filtering or amplifying excessively, which is not its job.
  - A noise analysis, which shows the circuit is not injecting severe noise of its own. Some noise is inevitable, but no noise density should be excessive.

***Optional Extra (Stretch Goals):***

- Plot the impedance, V(node)/I(source), as a function of frequency. Make sure that its impedance has no resonant valleys which would effect the measurement, and it provides a very large overall impedance.